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Call 2026 International collaboration in high-performance computing 01.04.2026 - 06.05.2026 23:59		Scientific Council RY	
Research topic/Project title Laaennettu moniskaalakoodikytkentä pölyn käsittelyyn supernova-aaltojen aiheuttamassa turbulenssissa Extended multi-scale code-coupling for dust processing in supernova-driven turbulence			
Funding period 01.01.2027 - 31.12.2029	Funding to be applied for 499 591 €	Overall cost estimate 713 702 €	
Keywords Astrofysiikka Numeeriset simulaatiot Magnetohydrodynamiikka Tähtienvälinen aine Turbulenssi Rinnakkaisohjelmointi Astrophysics Numerical simulations Magnetohydrodynamics Interstellar medium Turbulence Parallel programming			
Field of research 1. The Milky Way, galaxies and cosmology 2. Astronomy 3. Software engineering, operating systems, man-computer interaction 4. Solar, stellar and interstellar matter physics			
Abstract A common challenge in modelling astrophysical systems is that there is such a vast range in scale of both time and space which are relevant to the properties and evolution of most astrophysical objects that they cannot be combined together. This applies across the range from interplanetary physics to galaxy clusters. We plan to implement groundbreaking coupled multi-tiered, or extended multi scale simulations (EMS), bridging previously unattainable scales of separation. Extreme scales of separation are generic in astrophysics, where this innovative concept shall be applied to dust, but our new interactive control tools would have applicability in other disciplines proliferated by large and heterogeneous data. Cosmic dust is the fundamental building blocks for planets and stars, and are seeded by supernova explosions and subsequent processing by the interstellar medium, giving rise to new generations of stars. Supernova blast waves are also understood to destroy much greater quantities of dust already present in the ambient ISM. Estimates of dust abundances in early galaxies appear too high, given prevailing models of destruction rates and accumulation mechanisms. This has even been considered by some as a dust budget crisis. Our recent breakthrough revealed how cosmic dust can survive destruction by supernovae (SNe) more than previously thought. We showed that, relative to the commonly used uniform interstellar medium (ISM), including a realistic in-homogeneous and turbulent ISM can reduce 50% the amount of ambient dust destroyed during the lifetime of an SN remnant. We also found that including the Lorentz force of the turbulent magnetic field acting on the dust further halved these losses, a much greater effect than had been anticipated. Our magnetic field, generated naturally from SN driven-turbulence, has energy only 5% that of the observed ISM. To reach full strength requires large scale dynamo with much greater domain and up to 100 times longer integration times. Modelling dust requires much shorter scales. So we aim to use the large scale field inherited from our dynamo models to apply the dust processing in zoomed in regions at smaller and smaller scales adding relevant physics to understand the relation between dust abundances and star formation rate. The tools to manage multi-tiered Galaxy-ISM coupled simulations can be extended to other astrophysical topics and numerical modelling more generally.			

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Degrees			
University of Oulu	Doctor	Astronomy	01.09.1999
University of Oulu	Licentiate	Astronomy	01.09.1998
University of Oulu	Master's degree	Astronomy	01.12.1996
Adjunct professorships			
University of Helsinki	Astronomy		01.11.2008
Professorships			
Aalto University Associate professor (tenured)	Tietotekniikka	01.01.2020 -	
Aalto University Adjunct professor	Astroinformatiikka	01.06.2016 - 31.12.2020	
Max Planck Institute for Solar System Research W2 professorship, non-tenured	Aurinkotutkimus	01.06.2016 - 31.08.2021	

Public description of project

Nontechnical description in English

We shall try to understand how cosmic dust is created and destroyed by supernova explosions. Cosmic dust is the fundamental building block of planets and stars. It is composed of chemical elements, including the heaviest atoms that are forged in the heat of the explosions marking the death of the most massive stars which become supernovae. The vast bulk of space in galaxies is occupied by the interstellar medium (ISM), a diffuse magnetized gas of mainly atomic and molecular hydrogen but also dust. Our models have demonstrated the critical protection by ISM turbulence and its magnetic forces from the destructive power of supernova blast waves on cosmic dust. Understanding this protection, and mechanisms similar to it, in more detail could further alleviate disparities between theoretical estimates of dust destruction and observed dust abundances. We shall construct a powerful multi-tiered computational framework to enable us to span the vast scales between full galaxy and dust dynamics.

Nontechnical description in Finnish or Swedish

Pyrimme ymmärtämään, miten kosminen pöly syntyy ja tuhoutuu supernovaräjähdyksissä. Kosminen pöly on planeettojen ja tähtien perusrakennusaines; se koostuu alkuaineista, mukaan lukien raskaimmista atomeista, jotka takoutuvat räjähdysten kuumuudessa, kun kaikkein massiivisimmat tähdet kuolevat supernovina. Suurin osa galaksien avaruudesta on tähtienvälistä ainetta (ISM), harvaa magnetoitunutta kaasua, joka koostuu pääasiassa atomisesta ja molekulaarisesta vedystä, mutta myös pölystä. Mallimme ovat osoittaneet, että ISM:n turbulenssi ja sen magneettiset voimat tarjoavat ratkaisevaa suojaa supernovien iskuaaltojen tuhoavalta vaikutukselta pölyyn. Tämän suojavaikutuksen sekä vastaavien mekanismien tarkempi ymmärtäminen voisi edelleen vähentää eroja pölyn tuhoutumisen teoreettisten arvioiden ja havaittujen pölymäärien välillä. Rakennamme tehokkaan, monitasoisen laskennallisen viitekehyksen, jonka avulla voimme kattaa valtavat mittakaavat kokonaisista galakseista pölyn dynamiikkaan.

Salary of principal investigator

Justify in the field below why you include the PI's salary for project management and/or thesis supervision and/or research in the total project costs.

Describe the salary in the field below if the PI does not have an employment relationship for the duration of the funding period.

Collaborators

Project collaborators

Arif Babul (babul@uvic.ca)	Canada
University of Victoria	
Lars Mattsson (lars.mattsson@su.se)	Sweden
Stockholm University -SU	
Florian Kirchschlager (florian.kirchschlager@Ugent.be)	Belgium
University of Ghent	
Mordecai-Mark Mac Low (mordecai@amnh.org)	United States
American Museum of Natural History	

Mobility

Mobility plan

Name	Duration (months)	From	To
How mobility ties in with the project			
Researcher	2.0	Finland, Aalto University	Canada, University of Victoria
Postdoctoral Researcher			
Visit to Department of Astrophysics to work with Prof. Babul			
Arif Babul	2.0	Canada, University of Victoria	Finland, Aalto University
Professor			
Collaboration on analysis of simulations and galaxy structure			

Affiliations

National research infrastructures and/or ESFRI roadmap infrastructures

Finnish Grid and Cloud Infrastructure	FGCI
Partnership for Advanced Computing in Europe	PRACE
IT Centre for Science	CSC RI

Centre of Excellence

Flagship

Brief data management plan

Brief data management plan (max. 2500 characters). Read more <https://www.aka.fi/en/research-funding/apply-for-funding/how-to-apply-for-funding/az-index-of-application-guidelines/data-management-plan/data-management-plan/>

Any algorithms, software packages produced, and simulation setups, will be released as open source software via the GitHub repositories related to each HPC tool. The Pencil Code and PC-A version numbers shall be reported with the publications for reference and reproducibility.

During the project simulation data will be generated of order up to a few hundred terabytes. We shall consider hosting some critical advanced snapshots of more interesting full system states with our US collaborators at the AMNH data archive, but in general we shall backup reduced data via diagnostic and post-processing or on-the-fly processing which has been published or required for visualization of data on FAIRDATA or ALLAS where it has not been part of a publication.

Final peer-reviewed journal manuscripts and supplemental information, such as data tables for graphical information in manuscript figures and for statistically processed averages will be posted in the project digital repository. Authors will ensure their publishing agreement allows the paper to be posted to the arXiv; alternatively, the project website will provide a reference to the journal articles coupled with the supplemental information.

Research ethics

Ethics statement

Ethical permit	Not required
Have you been found guilty of a serious violation of research integrity during the last three years?	No
Guilty of illegal employment within the meaning referred to in section 7(2) of the Act on Discretionary Government Transfers (688/2001)	No

Progress report

Decision number	Project title	Call	Finance period
359462	Investigating COSmic MAGnetism with GPU-accelerated codes (COSMAGG)	Finansiering för mobilitetssamarbete med Kina, Taiwan eller Tyskland	01.04.2024 - 31.03.2026

Progress report

Immediately after the acceptance of the project on both sides, we started bi-weekly telecons between the project partners. In these telecons we have advanced the collaborative work remotely, but also the workplan and timetable of the collaborative visits. The first visit took place in June 2024 by doctoral student Johannes Pekkilä, to visit Taiwanese partners for a month. Work conducted and completed during the visit was the implementation of spherical coordinates and non-equidistant grids to the Astaroth library.

During February, 2025 3 colleagues from ASIAA visited Aalto for three weeks. We implemented additional physics relevant for the studies of the interstellar matter and other compressible fluids to the Astaroth library. Our doctoral student Touko Puro has visited ASIAA during summer 2025 to finalise these implementations.

Next round of visits took place in January-February by Touko Puro and Matthias Rheinhardt, who visited ASIAA for a month both. During this time, the first publication of the collaboration was submitted. Another publication will be submitted in due course. This project adds to the general numerical building blocks to the current application, but does not overlap with it in any other way.

Decision number 373416	Project title Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)	Call 2025 Special funding for international collaboration in high-performance computing	Finance period 01.01.2026 - 31.12.2028
Progress report We have obtained our first results with simulation-in-the-loop framework for forced helical turbulence and Taylor-Green vortex for hydrodynamic flows with U-Net and Fourier Neural Operator ML models. The agreement with the DNS and LES appears promising. One MSc thesis has been submitted about the results obtained during the project, and now we are writing the first peer-reviewed publication on the results.			

Most relevant publications and other key outputs

Publications				
Year of publication	Authors of publication	Name of publication	Name of journal/series	Permanent web address
2023	Gent, Frederick A.; Mac Low, Mordecai Mark; Korpi-Lagg, Maarit J.; Singh, Nishant K.	The Small-scale Dynamo in a Multiphase Supernova-driven Medium	Astrophysical Journal	
2024	Gent, Frederick A.; Mac Low, Mordecai-Mark; Korpi-Lagg, Maarit J.	Transition from small-scale to large-scale dynamo in a supernova-driven, multiphase medium	Astrophysical Journal	http://adsabs.harvard.edu/abs/2023arXiv230607051G
2024	Korpi-Lagg, Maarit; Mac Low, Mordecai-Mark; Gent, Frederick A.	Computational approaches to modeling dynamos in galaxies	Living Reviews in Computational Astrophysics	
2024	Kirschlager, Florian; Mattsson, Lars; Gent, Frederick	Supernova dust destruction in the magnetized turbulent ISM	Nature Communications	
2024	Nguyen, Minh Tri; Nguyen, Anh Dung; Rantaharju, Jarno; Puro, Touko; Rheinhardt, Matthias; Korpi-Lagg, Maarit; Truong, Hong Linh	Supporting Opportunistic Data Operations for Data-Intensive Computational Applications	IEEE International Conference on Big Data	
2025	Pekkila, Johannes; Lappi, Oskar; Robertsen, Fredrik; Korpi-Lagg, Maarit J.	Stencil Computations on AMD and Nvidia Graphics Processors: Performance and Tuning Strategies	Concurrency and computation: practice and experience	
2025	Gozaliasl, Ghassem; Yang, Lilan; Kartaltepe, Jeyhan S.; Toni, Greta; Abedini, Fatemeh; Akins, Hollis B.; Allen, Natalie; Arango-Toro, Rafael C.; Babul, Arif; Casey, Caitlin M.; Chartab, Nima; Drakos, Nicole E.; Faisst, Andreas L.; Finoguenov, Alexis; Flayhart, Carter; Franco, Maximilien; Ghaffari, Zohreh; Leroy, Gavin; Haghjoo, Aryana; Haghi, Hosein	Brightest group galaxies in COSMOS-Web : Evolution of the size-mass relation since $z = 3.7$	Astronomy and astrophysics	
Justification for selected publications				
The first three publications relate to the MHD turbulence simulations applying massively parallel HPC on both CPU and GPU platforms. The fourth publication highlights the breakthrough analysis of dust processing by supernova blast waves in turbulent magnetized plasma. The fifth and sixth publications highlight the development of the machine learning tools and GPU stencil code critical to the use of Lumi by this project. The last publication relates to the largest scale collaboration with our Canadian partner for coupling between Pencil Code and other codes.				
Other key outputs				
Output		Name of output		
patents and innovations		Astaroth		
developed practises, tools and software		Pencil Code		
Motivation for selected outputs				
Aalto HPCLab have developed the GPU sixth order stencil based PDE solver Astaroth (Invention disclosure IPID3713) and have implemented the integration of Pencil Code with the Astaroth PDE solver (F. A. Gent, M.-M. Mac Low, M. J. Korpi-Lagg, T. Puro, and M. Rheinhardt. Asymptotic Behaviour of Galactic Small-scale Dynamos at Modest Magnetic Prandtl Number. ApJL, 1000(2):L40, Apr. 2026. doi: 10.3847/2041-8213/ae4c41.)				

PROJECT FUNDING

Total costs

Salaries	2027	2028	2029	Total
Postgraduate student	36 300	37 026	37 763	111 089
Postdoctoral Researcher	53 900	54 978	56 078	164 956
Indirect employee costs 47.0 %	42 394	43 242	44 105	129 741
Overheads share 72.0 %	95 468	97 377	99 321	292 166
SALARIES TOTAL	228 062	232 623	237 267	697 952

Other costs (incl. VAT)	2027	2028	2029	Total
Services	100	100	100	300
Travel expenses	5 000	5 000	5 000	15 000
Other costs	150	150	150	450
OTHER EXPENCES TOTAL	5 250	5 250	5 250	15 750

Total project costs

Total	2027	2028	2029	Total
	233 312	237 873	242 517	713 702

Funding received from other sources

Funding sources	2027	2028	2029	Total
Own organisation	69 994	71 362	72 755	214 111
Total	69 994	71 362	72 755	214 111
Other funding (%)	30.00	30.00	30.00	30.00

Funding to be applied for from the Research Council

Total	2027	2028	2029	Total
	163 318	166 511	169 762	499 591
Research Council funding contribution (%)	70.00	70.00	70.00	70.00

Justification

Salary for Senior Researcher with expertise in the Project, Student to be recruited. Travel costs include research visits to and from Canada, travel between other collaborator host sites and expenses to attend international conferences for the student and host team. Other expenses include other modest sundry expenses that might arise.

Appendices

Heading	File	Date
CV	Korpi_Lagg_CV_3page.pdf	29.04.2026
Research security assessment and risk management plan	researchsecurity_RSf2026_Korpi_Lagg.pdf	05.05.2026
Complete list of publications	Korpi_Lagg_publ_wc.pdf	05.05.2026
Letter of collaboration	Letters_of_collaboration.pdf	06.05.2026

Extended multiscale code coupling for dust processing in supernova-driven turbulence

Maarit J. Korpi-Lagg (SNDust)

1 Background and significance

We plan to implement groundbreaking *extended multiscale simulations* (EMS), bridging previously unattainable separations of spatial scales. Extreme separation of scales is generic in astrophysical models. We shall apply this innovative concept to modeling the evolution of interstellar dust, but our new interactive control tools may apply to any disciplines characterized by large and heterogeneous data.

A common alternative approach to modeling multiple scales is *adaptive mesh refinement*, in which higher resolution meshes are dynamically applied locally where, typically, pressure gradients exceed some threshold (Berger and Colella, 1989). However, turbulent flows have intermittency everywhere, so refinement occurs throughout the domain for resolutions smaller than 1024^3 zones (Kritsuk et al., 2006). This does not readily allow adding small-scale physics only to regions where the smallest scales are resolved. Additionally, non-uniform discrete Fourier transforms that could be used to understand the spectral properties of turbulent fields are challenging to calculate (Greengard and Lee, 2004). Multigrid methods are another algorithm commonly employed for improving convergence with implicit solvers of highly non-linear systems in chemistry (Gerlinger et al., 1998), engineering and other disciplines, with increased overheads. Our approach will use multi-grid tools to extend the scales and physics accessible to the hierarchy of simulations with minimal increase in overheads.

An existing framework for applying a common interface to multiple astrophysics algorithms is the Astrophysical Multipurpose Software Environment (Pelupessy et al., 2013). Its Python interface is relatively user friendly, but has limited scalability via parallel high performance computation (HPC). Our project shall implement integrated, highly scalable simulations capable of handling huge data, with also user friendly interactive tools with which to manage and interrogate the applications.

Cosmic dust plays a critical role in many astrophysical processes, including as a catalyst in star formation, the main ingredient in planet formation, and, due to its light absorption and scattering, a contributor to the multiphase structure of the interstellar medium (ISM), galaxy evolution, and the properties of cooling in the circumgalactic medium in cosmological scale simulations (Saeedzadeh et al., 2023). Heavy elements are forged in stars, the heaviest metals fusing at the explosive death of the largest stars, and thus ejected into the ISM by supernovae (SNe). Some of these form dust, comprising a mixture of carbonaceous material and silicates (Weingartner and Draine, 2001). The abundance or composition of dust is indicative of a galaxy's age and star formation rate. Although SNe eject into the ISM some fraction of dust, their highly supersonic blast waves can destroy dust already present in the ambient ISM (McKee, 1989). Observations of dust abundances in early galaxies substantially exceed predictions from prevailing models of destruction rates and accumulation mechanisms. This has been described as a **dust budget crisis** (Rowlands et al., 2014). Sublimation, condensation, grain-grain interactions, and ion sputtering are well established as primary dust destruction mechanisms in SN blast waves (Jones and Nuth, 2011). The resultant dust properties are, however, highly uncertain. Both processing and formation of ISM dust remain poorly understood (Draine, 2009) and most models omit realistic ISM magnetized turbulence.

Our recent breakthrough models demonstrate that ISM dust can survive destruction by SN blast waves far better than previously thought (Kirchschlager et al., 2024). We show that, compared to the commonly used uniform ISM, including a realistic inhomogeneous and turbulent ISM can reduce by 50% the amount of ambient dust destroyed during the lifetime of an SN remnant. We

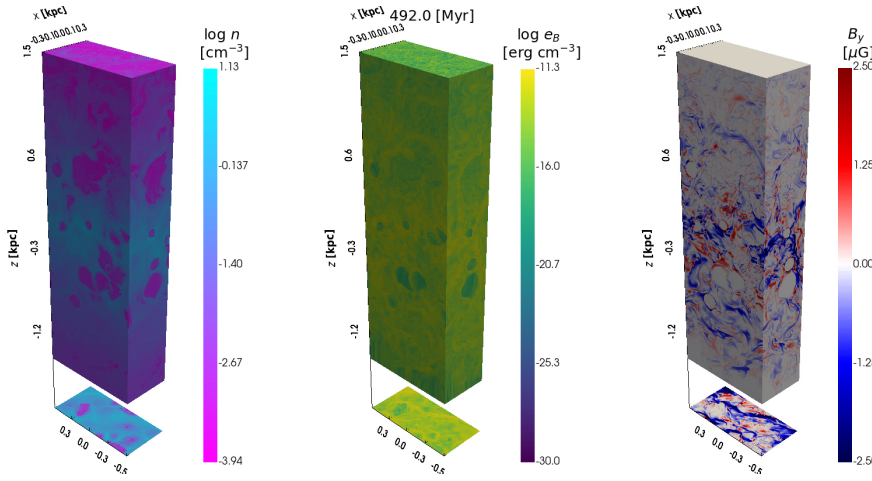


Figure 1: Typical snapshots of gas number density n , magnetic energy density e_B , and azimuthal magnetic field B_y from an MHD simulation of SN-driven turbulence and galactic dynamo activity (Gent et al., 2024). Turbulence drives a fast small-scale dynamo, which generates magnetic fields up to a few percent of the kinetic energy of the turbulence, while stratification about the galactic disc regulates the multi-phase structure of the ISM via outflows and inward precipitation. Large-scale shear is required for a mean-field dynamo, which continues the growth of the magnetic field to energy equipartition with the kinetic energy.

also find that including the Lorentz force of the turbulent magnetic field acting on the dust further halves these losses, a much greater effect than had been anticipated. Even larger scales are required to fully model the magnetic field structure and yet smaller scales to properly understand the dust processing. To advance this scientific goal we need our proposed EMS approach and machine-learning toolkit to facilitate identifying critical features, launching coupled simulations, managing communication and balancing resources within a typical HPC batch system.

Our existing simulations, performed to sixth-order spatial accuracy, explore many scales of the ISM. Over such vast ranges of scales in space and time, all models, whether of full galaxy or star-forming molecular clouds (MC), must exclude some physics from their scope. At kiloparsec scale (see, e.g., Figure 1) SN-driven turbulence in stratified discs with galactic-shear dynamos (e.g., Gent et al., 2024) neglects dust processing, but acquires a necessary turbulent structure. Dust decouples from the gas (Mattsson et al., 2019) and radiation pressure (Heinemann et al., 2006) and Lorentz forces (Draine and Salpeter, 1979) affect the critical distribution of grains, easily charged, at MC scale.

To bridge the gap between scales relevant to dust processing and the numerical scales required to model the realistic multi-phase turbulent ISM, we have treated dust as a distribution of multiple fluids binned by grain size and applied via the post-processing tool PAPERBOATS (Kirchschlager et al., 2019) to prior MHD data. Kirchschlager et al. (2024) applied PAPERBOATS to process dust on 2D slices taken from 3D MHD models of the turbulent SN-driven dynamo (Gent et al., 2021, 2023) obtained with PENCIL CODE (Pencil Code Collaboration et al., 2021). As illustrated in Figure 2 the magnetic field has a significant effect on dust resilience, even beyond the effects of turbulence. A comprehensive review of MHD simulations at multiple scales (Whitworth et al., 2025), exploring correlations between gas number density n and magnetic field strength B , concludes that Gent et al. (2024) obtained $B - n$ relations that are among the best match to those from observations. Large scale dynamo (LSD) and a realistic multi-phase ISM are thus critical ingredients.

We seek to quantify the rate at which dust is processed by SN blast waves: how this depends on the multiphase turbulence of the ISM and on the strength and topology of the galactic magnetic field. By including realistic large scale fields and large scale galactic topology in increasingly high resolution zoom-ins of the ISM we shall clarify where and when blast waves are most destructive and what processes promote or mitigate dust loss in different phases of the ISM.

Our primary tool for massively parallel GPU computations is the sixth-order MHD code PENCIL, as accelerated by the ASTAROTH framework (Pekkilä et al., 2022), hereafter referred to as PC-A, developed at [HPCLab](#) (High Performance Computing Laboratory) in the Department of Computer

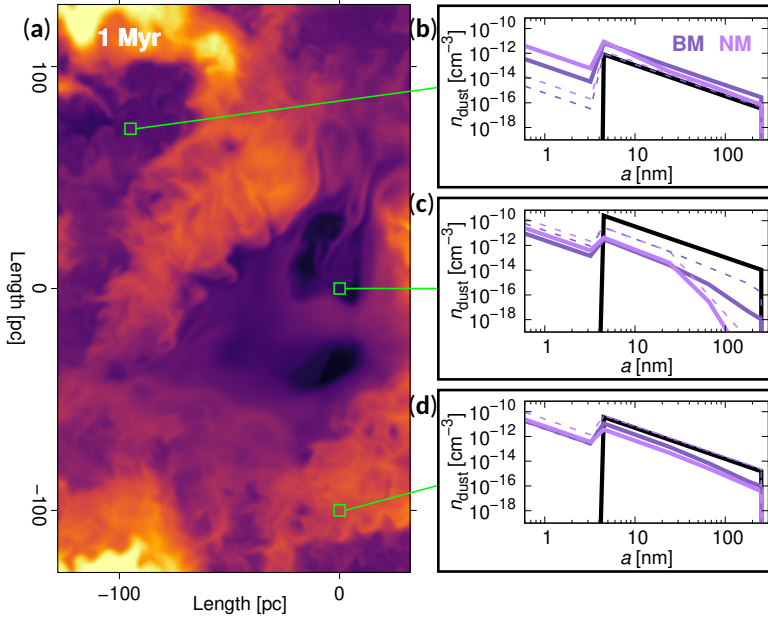


Figure 2: Regional variation of dust distribution at sampled times. (a) Location of three specimen regions (green boxes) in the vicinity of an SN blast wave, each of size of $5 \text{ pc} \times 5 \text{ pc}$. Dust distributions (c) about the SN center, and (b) and (d) at unshocked diffuse or moderate density regions, respectively. Each depicts distributions: initially MRN (Mathis et al., 1977, black), after 200 kyr (dashed lines) and after 1 Myr (solid lines). BM and NM indicate with and without Lorentz force, respectively. The dust distributions show the average density for each grain size in each specimen. At region (c) significant amount of dust grains are removed or destroyed by the shock while at regions (b) and (d) any change is due to background processing (transport and destruction).

Science (hereafter Aalto/CS) at the Aalto University School of Science. The project suitability for HPC international collaboration has been demonstrated by successful campaigns at LUMI Extreme Scale 2025 (Gent et al., 2026) and FRONTIER (USA) [INCITE 2026](#) now in progress. Adding dust processing, at low Stokes numbers requires higher time resolution, and large-scale dynamos, requires greater domains and duration. Acceleration with GPUs on LUMI is critical to the feasibility of such studies. PC-A has attained over $20\times$ speed-up over PENCIL CODE on CPUs alone. It has performed reliably and stably with weak scaling approaching near constant execution time at up to 4096 GPUs on FRONTIER.

2 Impact

It is a primary focus of this project to bring together two or more codes, or instances of the same code, working on related but distinct physics. The participants will develop internationally leading expertise in knowledge and application of our collaborative codes and how to create, adapt and apply machinery to support their synthesis.

The critical advance to application science envisaged here is to model physics, previously mutually exclusive due to differences in scale, but essential to gas-dust and dust-dust dynamics:

- *Kinetic drag*: Epstein-type, gas-dust interaction, significant at typical MC scales (10–20 pc).
- *Coulomb forces*: Electrically charged dust affected in a plasma.
- *Lorentz forces*: Dynamo-generated magnetic fields exerting forces on charged grains.
- *Self-gravity*: Cold cloud high densities act on decoupled grains, affecting dust clustering.
- *Radiation pressure*: Monochromatic proxy prior to SN explosions.
- *Dust cooling*: Supersede gas temperature-dependent cooling functions in MHD simulations.

The scientific impact of the project shall be to extend our understanding of the processing of cosmic dust by supernovae and to accurately relate cosmic dust abundances to star formation rates. The comprehensive treatment of dust processing available through PAPERBOATS and its incorporation directly into PENCIL CODE and PC-A applied to realistic multiphase ISM turbulence provides a more accurate statistical correlation between various dust grain sizes and gas abundances. From this we propose to provide the most reliable model for dust composition and distribution in galaxies

pertaining to its age, position within the galaxy structure and its magnetic field topology.

Effective implementation of the EMS approach will replace manual inspection of data, archived to disk at run completion, with machine learning tools using idle processors to access the data in memory at run time. Current strategies identify a maximum density and zoom in around its neighbourhood, or apply algorithms to identify clusters in which star formation may be expected within a time frame determined by its freefall time. New highly scalable and efficient algorithms shall inspect the data while simulations continue to advance. Subdomain extraction, criteria for required physics, resolution and other simulation parameters with subsequent execution need also to become automated. We plan to run the simulations at different resolution or with complementary physics simultaneously, developing tools to manage communication across the nodes and distribute resources efficiently between the simulations, potentially extending to a hierarchy of simultaneous EMS.

As the hierarchy of subdomains expands, storage becomes an increasing bottleneck, making analysis and interpretation more challenging. An objective is to develop an intelligent adaptive **pipeline** for managing the implementation, evolution and measurement of simulations. As proof of concept this applies to our astrophysical simulations, but ultimately any HPC applications with large highly heterogeneous data. We shall develop code coupling products to combine calculations of related problems with very different methods or regimes and data management products to diagnose and adapt HPC applications on-the-fly, employing machine learning to recognize critical features, and issue commands using **opportunistic data operations** (ODOP) (Nguyen et al., 2024).

Critical advances to the HPC toolkit:

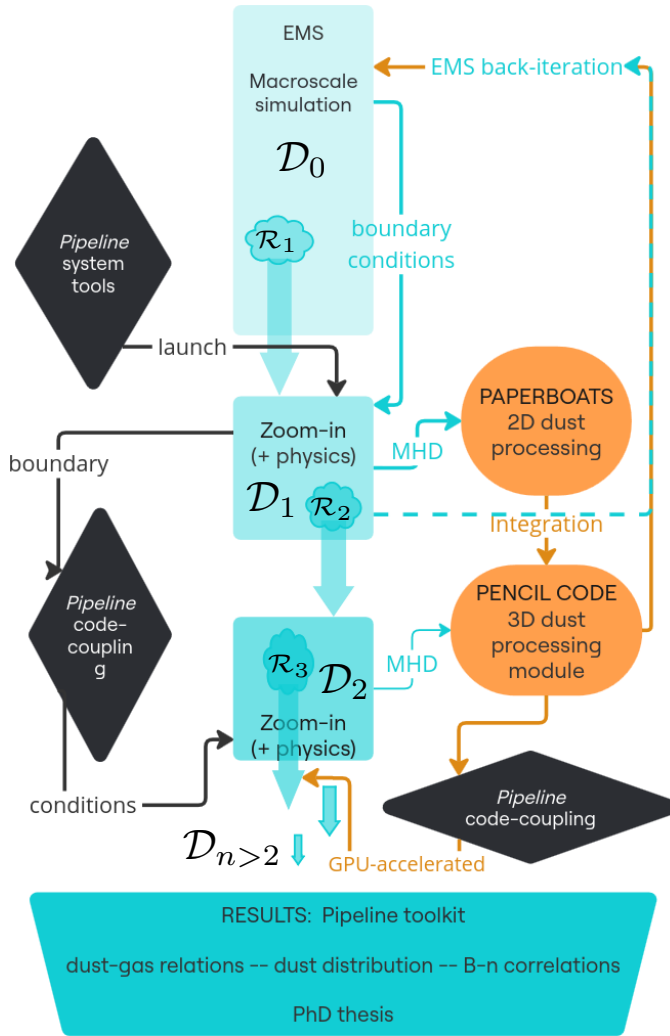
- *Feature detection*: 3D cloud cluster, gravity hotspot, user-specified feature detection tools.
- *Subdomain launch*: Automated choice of subdomains, denoted $\mathcal{D}_n$ (see below section 3.1), new physics and parameter setting, compilation and execution.
- *Code coupling*: Adaptive resource and communication management of coupled simulations.
- *On-the-fly-diagnostics*: Running diagnostics concurrently with the simulations (using ODOP).
- *Node management*: Concurrent computation, memory transfers, and diagnostics.
- *GPU acceleration*: Dust processing, radiation, and self-gravity added to MHD on GPUs.

3 Implementation

3.1 Work plan and schedule

WP1 (EMS): Extended Multiscale Simulations

From established *macro-scale simulations*, in which micro-scale physics of gas-dust and dust-dust interactions are neglected or treated in PC-A with a mean dust density or dust velocity approach, but using stratification and shear we spawn consecutive EMS from a zoomed-in region of the larger domain (see flow chart). We capture scales between sub-MC sizes and the large-scale MHD flow phenomena via *boundary conditions* (BCs) for the subdomain inherited and interpolated from the larger host domain in which it is embedded. The span between macro and micro scales is too large to bridge in one step. With each increase in resolution additional physics can be considered, such as self-gravity (Krasnopolsky et al., 2026, recently added to PC-A) or stellar radiation and, eventually, dust. EMS is not an entirely new concept (Xia et al., 2023; Carrasco-Busturia et al., 2024), but offers an innovative means to perform the required simulations for our science goals. Applying such machinery alongside other codes, we aim to embed our largest-scale dynamo models in the galaxy scale models of our Victoria collaborators using Arepo or ChaNGa (Lim et al., 2021; Oppenheimer et al., 2021), to capture the full structure of the ISM magnetic field. Selecting suitable subdomains within such full-galaxy simulations, we shall map critical variables from the unstructured Arepo or



Schematically, the EMS process is as follows:

1. Select a suitable macro simulation on domain $\mathcal{D}_0$.
2. Identify region $\mathcal{R}_n$ (e.g., dense star-forming cloud) within $\mathcal{D}_{n-1}$, small enough to ensure scale separation.
3. Extract domain $\mathcal{D}_n$ enclosing $\mathcal{R}_n$ and remesh to the desired resolution as initial conditions (ICs).
4. Define appropriate boundary conditions (BCs). Interpolate from $\mathcal{D}_{n-1}$ to retain macro dynamics.
5. Run simulation of $\mathcal{D}_n$.
6. Repeat steps 2-5 for $n > 1$ until the size of $\mathcal{D}_n$ reaches some limiting length scale.

ChaNGa subdomain onto a matching structured coarse grid for PC-A. This shall be updated as required to host the higher resolution multiphase ISM dynamo simulations.

Deliverables and expected results:

- MHD simulations spanning more than three tiers or two orders of magnitude in resolution.
- Models of gas-dust interaction and B - n correlations in ISM environments up to MC densities.
- Hand-setting $\mathcal{D}_n$ & boundary conditions superseded by pipeline tools (see **WP3**).
- Post-processed dust (PAPERBOATS) superseded by concurrent MHD and dust evolution by the GPU-accelerated PENCIL CODE (PC-A) (**WP2**).

WP2 (Dust): Dust processing

PAPERBOATS shall be applied to new *multiscale* simulations over a wider range of ISM environments, while being fully implemented into the efficiently parallel, sixth-order PENCIL CODE by updating its dust modules to include the full functionality of PAPERBOATS. Due to transposition of the Fortran code this will be inherited by the GPU-accelerated PC-A. Code coupling of MHD and dust modeling will be developed to address differences in timescales and computational load required for faster and more efficient deployment of resources (see **WP3**).

Deliveries and expected results:

- PAPERBOATS implemented into highly scalable PC-A.
- Dust modeling applied with PC-A.
- Enhanced modeling of dust to include radiation pressure and self-gravity.
- Enhanced modeling of cooling to based on dust.
- Code-coupling tools: highly heterogeneous codes or computations require resource management

to avoid idle resources, match performance and reduce IO, memory or compute bottlenecks (**WP3**).

WP3 (CC): Pipeline – code coupling

Defining consistent BCs after remeshing to some micro-scale $\mathcal{D}_n$, without erroneous artifacts in the flows, is a crucial step in the process. Open boundaries independent of $\mathcal{D}_{n-1}$ can work (Copabianco, 2025, and paper in preparation), but acquire only initial features from $\mathcal{D}_{n-1}$, which dissipate over time. Using surfaces from within the higher tier domain $\mathcal{D}_{n-1}$ surrounding $\mathcal{D}_n$ one may interpolate boundary values onto the fine grid (also in time) to retain dynamically the macro-scale features. High cadence slices from $\mathcal{D}_{n-1}$ can be saved to disk and read subsequently for $\mathcal{D}_n$ BC updates.

Data storage and management is commonly a bottleneck in any HPC, requiring management of datasets approaching TB and even PB scales. Lossless data compression for such data can provide modest storage reductions, but code coupling on-the-fly saves substantially more. It employs at the very least message passing interface (MPI). Additional physics, such as dust, cooling or self-gravity, necessitate diverging timescales or computational load. Code coupling enhanced by machine learning to minimize redundant communication could gain speed, balancing load and available resources. We have access to cutting-edge machine-learning tools to inspect the data for regions or patterns of interest and to extract as a suitable subdomain from which to launch new experiments. This work is at the frontier of computer science, in which the Aalto/CS HPCLab and [AaltoSEA](#) are heavily invested.

The basic machinery would be similar for coupling multiple instances of the same code (e.g. PC-A) or with other codes. HPCLab already has experience coupling [Eulag-MHD](#) (Monteiro, 2024) with PENCIL CODE (1-way). In this case, the evolving data from the highly stratified solar atmosphere obtained through a Eulag-MHD simulation was passed on-the-fly to the PENCIL CODE to apply the test-field method to calculate the turbulent transport coefficients that can be used to explain the dynamo.

Deliveries and expected results:

- Code coupling to communicate BCs **WP1** across multiple scales (1-way).
- Code coupling to share code with varying physics, e.g. dust processing with MHD **WP2**, and to balance computational load (2-way).
- Code coupling to communicate data between CPU-based simulations and GPU-simulations (2-way).
- Code coupling to combine simulations using different codes (1 or 2-way).

WP3 shall provide the framework around which the student dissertation shall develop. Early in **WP3**, the student shall understand what manual interaction is applied to the selection and management of data for **WP1** and develop machine management tools to automate and learn these tasks, including where applicable the use of ODOP. The student with the support of the PI and Col will design and test new code coupling machinery. Initial methods could rely on manual settings at execution, but subsequently the aim would be to extend to more automated control and responses to the coupled codes to maximize utilization and reduce compute times.

3.2 Research data and material, methods and research environment

The simulation data is mainly generated in HDF5 form for field data of TB scale and ASCII for integrated quantities over time. Astrophysical observational data may be analyzed alongside our generated data for scientific publication. We also produce and utilize open source numerical codes and additional open source software tools for code integrations, management pipelines, and data analysis. The primary code milestones of this project will include: the implementation of PAPERBOATS' advanced dust processing physics into PC-A for accelerated parallel 3D integration;

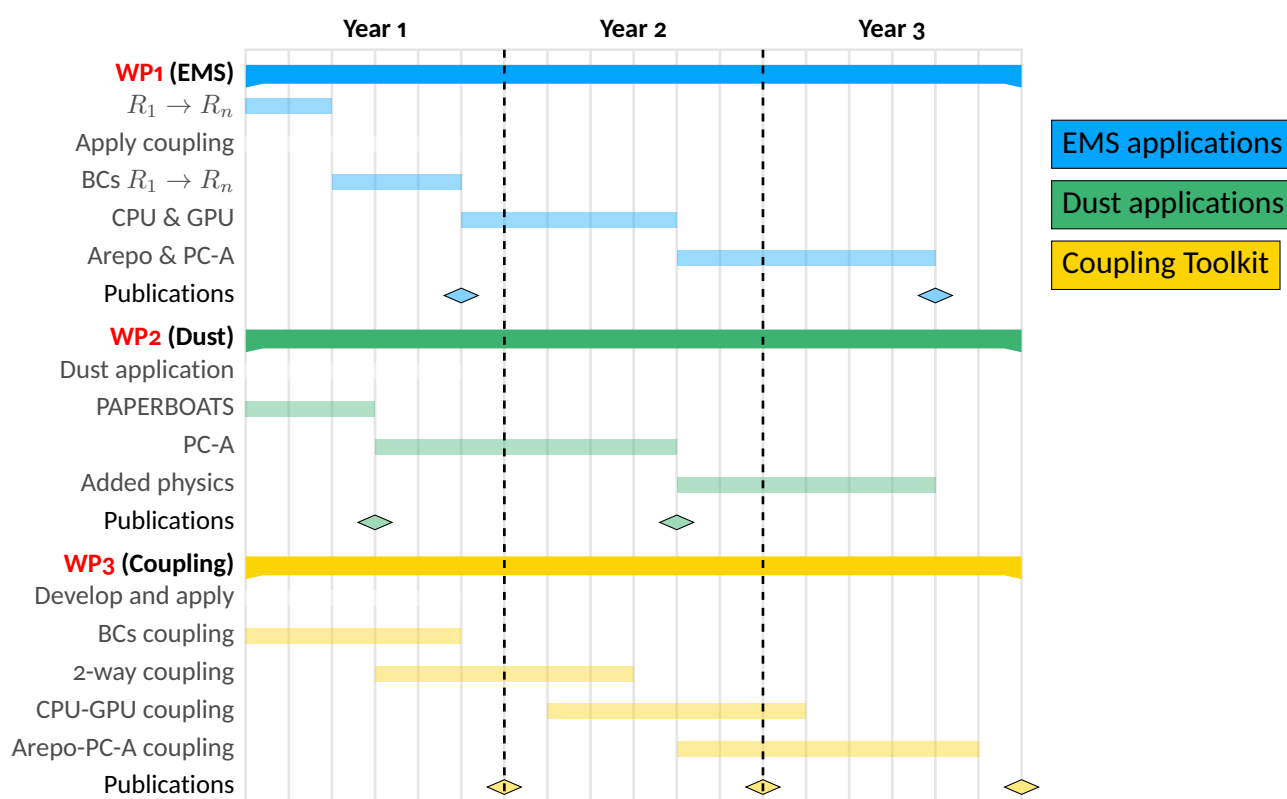


Figure 3: Gantt chart: proposed 3-year timeline and **WP** organization. Blue denotes simulations within EMS with coupled PC-A instances and with other codes; green denotes development and application of dust modeling; yellow denotes development and application of coupling strategies. At least 3 publications shall have the student as lead author reporting code and tool development or results of applications of these new tools; yellow to [Nature Machine Intelligence](#) or [IEEE Transactions on Automatic Control](#); blue and green to [Nature Astronomy](#) or [Astronomy & Astrophysics](#).

code coupling machinery for running multiple instances of PC-A or to match difficult combinations of scales and physics or other codes; automated code coupling and simulation management support.

The main research environment Aalto/CS. With over 40 professors and more than 400 employees, the department is the largest Computer Science unit in Finland. It is ranked among the top 10 Computer Science departments in Europe and among the top 100 globally. Information and communications technology and digitalization are among the seven key research areas at Aalto/CS. In addition to standard resources provided by the department, there is support for midrange HPC and data analytics through the Science-IT project of the Aalto University School of Science. Science-IT and its flagship HPC unit Triton, are a part of the national Finnish Computing Competence Infrastructure (FCCI). Currently, Triton provides 9600 computing cores and over 30 servers for high-end GPU computing. For in-depth research methods and software development, Science-IT provides support from research software engineers. Researchers also have free access to national CSC resources, including the supercomputers PUHTI and MAHTI, as well as the cloud resources Rahti and CPouta. The teams also have access to Europe's most powerful GPU-based supercomputer LUMI through competitive computing time allocations via the Euro-HPC consortium. Our collaboration has an outstanding track record of obtaining competitive computational resources through the Partnership for Advanced Computing in Europe (PRACE), EuroHPC, and CSC.

3.3 High-performance computing resources

The project will rely heavily on massively parallel GPU simulations. As coupling codes designed for use on varying architecture is an important element of the project we shall also seek massively parallel CPU resources. Running times for typical jobs often require walltimes in excess of a few weeks, due to the long integration times to evolve galactic dynamos, an important aspect of our science objectives. Small-scale dynamo (SSD) simulations on LUMI and FRONTIER each have typical walltime above two weeks, the duration differing little by model resolution, due to the highly efficient scaling of the PC-A. The time step reducing with resolution is offset by the increasing efficiency of the dynamo. Dust processing typically drives down the timestep while the duration of the simulations would not need to be as long, so resources for these simulations we would expect to be similar to those of the SSD. We anticipate the need for multiple millions of GPU node hours and plan to participate in Finnish, European, Canadian and US extreme scale calls at every opportunity.

3.4 Risk assessment and alternative implementation strategies

Some risk to **WP1** arises from delays or challenges in developing the code coupling in **WP3**. An alternative strategy remains to manually select regions of interest on which to post-process dust. For embedding the PC-A large-scale dynamo in the galaxy data of Arepo or ChaNGa we could manually map the data to the grid. Open BCs could be used on subdomains should interpolating these from the macro-scale prove unfeasible. This has already been done for a smaller domain (Copabianco, 2025, and paper in preparation). If the boundaries are distant from the region of interest and the duration is short this can be effective.

Modifications to the dust modules and implementation to PC-A **WP2** are low risk. The PI and Col have led the HPCLab development of the PC-A implementation of SN driven turbulence. Partial dust-processing modeling has successfully been implemented by the Lars Mattsson. Florian Kirchschrager, author of PAPERBOATS, can advise on any technical challenges. Further extension of the physics of dust modeling depends on the successful implementation of **WP1** and **WP3** to enable ever higher resolution domains to be investigated.

For **WP1** and **WP2** the PI and Col have years of consistent success applying for resources on the Finnish national infrastructure CSC, European infrastructure directly on LUMI, as well as successful HPCLab European scale PRACE projects. The Col and Lars Mattsson also have access to the National Academic Infrastructure for Supercomputing in Sweden. Via Mordecai Mac Low we can apply for FRONTIER HPC resources. All collaborators have high expertise at modeling dust and MCs. However, with fewer simulations we would still have access to scientific breakthrough and enough for testing and running of the new code coupling tools in **WP3** to enable the success of the student. **WP1** and **WP2** are assured results of high impact, since they explore regimes of astrophysics with respect to dust distribution and dynamics, magnetic field and ISM multi-phase structure that are poorly understood and important to interpreting observations and understanding galaxy evolution. Four publications are realistic, the rate and extent of any advances increasing the better the success of **WP3**. The graduate student shall be included in the execution and analysis of **WP1** and **WP2**, while their thesis shall primarily address **WP3**.

Risks associated with code coupling in **WP3** include effectively managing communication between asymmetric processes in the MPI distribution of the different simulations and effectively managing the load between the nodes in the pair to obtain optimal load and maximum speed. We have had experience at handling this but if necessary compromises can be made in domain selection to facilitate processor communication at the expense of a slightly larger grid. Most of the coupling is between instances of PC-A, making it easier to share domain data. In the case of other codes, suitable intermediary mapping of the key data can be implemented to prepare it for easier transfer between codes. If the progress of **WP3** is delayed, manual management of the EMS process can still

permit us to make progress with the scientific objectives of **WP1** and **WP2**.

3.5 Project personnel and their project-relevant key merits

The Principal Investigator (PI) Maarit Korpi-Lagg^{} (over 4,800 citations and h-index of 39 according to Google Scholar), is a tenured associate professor at Aalto/CS, where she leads the High-Performance Computing Lab. She was the grant holder of the ERC Consolidator grant "UniSDyn" and the ERC Proof-of-concept grant "SYCOS", projects that studied dynamo action in solar and stellar convection zones. She acted as a team leader of the Academy of Finland Centre of Excellence "ReSoLVE" (2014-2019), which studied long-term solar magnetic activity and its effects on Earth. She has acted as a PI of several competitive HPC projects (LUMI-C and LUMI-G Finnish project allocations, PRACE, DEISA/DECI, CSC Grand Challenge), and is a core developer of the PENCIL CODE Multiphysics Solver for high-order stencil-based applications, and the initiator and project leader of the GPU-based MHD solver ASTAROTH. The PI shall have responsibility for recruitment and supervision of the Graduate Student (GS) and the integration of **WP3** within the HPCLab team.

The Co-Investigator (CoI) Frederick Gent^{} (1000 citations and h-index of 17) was a Research Fellow at Aalto/CS until March 2025 and since November 2023 has been employed as Assistant Professor at Nordita, Stockholm, under the Swedish Research Council grant no. 2022-03767. The CoI is a core developer of the PENCIL CODE and has the first published use-application of PC-A (Gent et al., 2026). The CoI will be the thesis advisor for the GS working on **WP1 – WP3** and implement **WP1** with Arif Babul & Mordecai-M. Mac Low and **WP2** with Lars Mattsson & Florian Kirchschrager. The CoI has managed three Grand Challenge projects around galactic large-scale dynamos on CSC since 2009, and on LUMI he led a 2025 Extreme Scale project, then running on FRONTIER (USA) a 2026 Incite project, both relating to the galactic SSD.

The Graduate Student (GS) shall be recruited to the Aalto/CS Graduate Programme. They will work on **WP3** and as lead author on three papers on code coupling or its application. The applications of the code development and tools shall be applied to at least one EMS application **WP1** and they will be included in the experimental design, implementation and publication of **WP1** and **WP2** as part of their training and education.

Arif Babul, University of Victoria, Canada^{} Distinguished Professor in galaxy evolution and composition. Arif has expertise in numerical modeling of galaxy evolution and shall contribute to **WP1**, providing subdomain data from galaxy formation simulations for galactic dynamo models and coauthoring publication of MHD **WP1** and dust **WP2** simulation results. He will also support the student in **WP3** on how to couple alternative codes with PC-A.

Mordecai-M Mac Low, American Museum of Natural History, New York, USA^{} Curator and Professor of Astrophysics with expertise of numerical modeling, astrophysical turbulence and molecular clouds. We have an ongoing collaboration resulting in several coauthored publications and sharing multiple HPC projects on CSC, LUMI and FRONTIER together including parallel runs as large as 16384 GPUs on FRONTIER. Mordecai will contribute to the experimental design and subsequent analysis of **WP1** and **WP2**.

Lars Mattsson, Nordita, Stockholm University, Sweden^{} Assistant Professor of Astrophysics and expert on dust processing in astrophysical turbulence. Lars is a core developer of the PENCIL CODE and worked with the CoI on modeling dust in high Mach number turbulence. Lars will contribute to **WP1** with the EMS design, analysis of **WP2** results and implementing dust with PC-A.

Florian Kirchschrager, Ghent University, Belgium^{} Heisenberg Fellow recently awarded at Hamburg University, Florian is an expert on cosmic dust processing, author of the PAPERBOATS Code and a regular collaborator of the CoI. Florian will assist the implementation of PAPERBOATS functionality into PC-A dust modules and coauthor publications on **WP2**.

4 Responsible science

There are no ethical concerns as the application deals with no sensitive data. Nevertheless, we will follow stringent integrity principles in accordance with the guidelines of the Finnish Advisory Board on Research Integrity, and the ALLEA European Code of Conduct for Research Integrity. We are committed to promote diversity, equality and non-discrimination in all our activities. Statutory obligations set forth in the Act on Equality between Women and Men 609/1986, 1329/2014 and the Non-discrimination Act 1325/2014 also apply to our plan. We are firmly committed to promote equal opportunities for individuals to learn, acquire knowledge, participate and make a difference. We will make our values visible in all our practices.

The products of the research will be made available immediately to the community during the project. Fully open access publications, data, and source code policy will be adopted. Services like Zenodo and FAIRdata IDA will be utilised for opening up data products ranging from the data to reproduce all the figures, to intermediate and selected raw data products for reproducibility and reusability of the results. All the simulation and data analysis tools are developed as open source, available and continuously maintained on GitHub and Bitbucket repositories.

Our simulation and data analysis tools are being developed to work generally on GPUs and hence are energy-efficient in comparison to standard CPU computing. Data-efficient methods in our work help to achieve sustainable storage (also avoid data waste) and energy efficiency. Furthermore, we assess if international and national travel can be avoided by using remote visits. Computational time will be used wisely to avoid too large energy consumption.

References

- M. J. Berger and P. Colella. Local Adaptive Mesh Refinement for Shock Hydrodynamics. *Journal of Computational Physics*, 82(1):64–84, May 1989. doi: 10.1016/0021-9991(89)90035-1.
- D. Carrasco-Busturia, E. Ippoliti, S. Meloni, U. Rothlisberger, and J. M. H. Olsen. Multiscale biomolecular simulations in the exascale era. *Current Opinion in Structural Biology*, 86:102821, 2024. ISSN 0959-440X. doi: [HTTPS://DOI.ORG/10.1016/J.SBI.2024.102821](https://doi.org/10.1016/j.sbi.2024.102821). URL [HTTPS://WWW.SCIENCEDIRECT.COM/SCIENCE/ARTICLE/PII/S0959440X24000484](https://www.sciencedirect.com/science/article/pii/S0959440X24000484).
- A. Copabianco. *Supernova-induced processing of interstellar dust in a turbulent, magnetized and high-density ISM*. Master thesis, University of Bath, May 2025. URL [HTTPS://AURORA-CAPOBIANCO.GITHUB.IO/DOCS/MASTER_THESIS_AC.PDF](https://AURORA-CAPOBIANCO.GITHUB.IO/DOCS/MASTER_THESIS_AC.PDF).
- B. T. Draine. Interstellar Dust Models and Evolutionary Implications. In T. Henning, E. Grün, and J. Steinacker, editors, *Cosmic Dust - Near and Far*, volume 414 of *Astronomical Society of the Pacific Conference Series*, page 453, Dec. 2009. doi: 10.48550/ARXIV.0903.1658.
- B. T. Draine and E. E. Salpeter. Destruction mechanisms for interstellar dust. *ApJ*, 231:438–455, July 1979. doi: 10.1086/157206.
- F. A. Gent, M.-M. Mac Low, M. J. Käpylä, and N. K. Singh. Small-scale Dynamo in Supernova-driven Interstellar Turbulence. *ApJ*, 910(2):L15, Apr. 2021. doi: 10.3847/2041-8213/ABED59.
- F. A. Gent, M.-M. Mac Low, M. J. Korpi-Lagg, and N. K. Singh. The Small-scale Dynamo in a Multiphase Supernova-driven Medium. *ApJ*, 943(2):176, Feb. 2023. doi: 10.3847/1538-4357/ACAC20.
- F. A. Gent, M.-M. Mac Low, and M. J. Korpi-Lagg. Transition from Small-scale to Large-scale Dynamo in a Supernova-driven, Multiphase Medium. *ApJ*, 961(1):7, Jan. 2024. doi: 10.3847/1538-4357/AD0DA0.

- F. A. Gent, M.-M. Mac Low, M. J. Korpi-Lagg, T. Puro, and M. Rheinhardt. Asymptotic Behaviour of Galactic Small-scale Dynamos at Modest Magnetic Prandtl Number. *ApJ*, 1000(2):L40, Apr. 2026. doi: 10.3847/2041-8213/AE4C41.
- P. Gerlinger, P. Stoll, and D. Brüggemann. An implicit multigrid method for the simulation of chemically reacting flows. *Journal of Computational Physics*, 146(1):322–345, 1998. ISSN 0021-9991. doi: [HTTPS://DOI.ORG/10.1006/JCPH.1998.6061](https://doi.org/10.1006/JCPH.1998.6061). URL [HTTPS://WWW.SCIENCEDIRECT.COM/SCIENCE/ARTICLE/PII/S0021999198960612](https://www.sciencedirect.com/science/article/pii/S0021999198960612).
- L. Greengard and J.-Y. Lee. Accelerating the nonuniform fast fourier transform. *SIAM Review*, 46(3): 443–454, 2004. doi: 10.1137/S003614450343200X. URL [HTTPS://DOI.ORG/10.1137/S003614450343200X](https://doi.org/10.1137/S003614450343200X).
- T. Heinemann, W. Dobler, Å. Nordlund, and A. Brandenburg. Radiative transfer in decomposed domains. *A&A*, 448(2):731–737, Mar. 2006. doi: 10.1051/0004-6361/20053120.
- A. P. Jones and J. A. Nuth. Dust destruction in the ISM: a re-evaluation of dust lifetimes. *A&A*, 530: A44, June 2011. doi: 10.1051/0004-6361/201014440.
- F. Kirchschrager, F. D. Schmidt, M. J. Barlow, E. L. Fogerty, A. Bevan, and F. D. Priestley. Dust survival rates in clumps passing through the Cas A reverse shock - I. Results for a range of clump densities. *MNRAS*, 489(4):4465–4496, Nov. 2019. doi: 10.1093/MNRAS/STZ2399.
- F. Kirchschrager, L. Mattsson, and F. A. Gent. Supernova dust destruction in the magnetized turbulent ISM. *Nature Communications*, 15:1841, Feb. 2024. doi: 10.1038/s41467-024-45962-0.
- R. Krasnopolsky, T. Puro, W.-W. Li, H. Shang, M. S. Väisälä, M.-M. Mac Low, M. Rheinhardt, and M. Korpi-Lagg. Iterative poisson solvers for self-gravity with the gpu code astaroth. *submitted to PASP*, 2026. URL [HTTP://ARXIV.ORG/ABS/2605.03563](http://arxiv.org/abs/2605.03563).
- A. G. Kritsuk, M. L. Norman, and P. Padoan. Adaptive mesh refinement for supersonic molecular cloud turbulence. *The Astrophysical Journal*, 638(1):L25, jan 2006. doi: 10.1086/500688. URL [HTTPS://DOI.ORG/10.1086/500688](https://doi.org/10.1086/500688).
- S. Lim, D. Scott, A. Babul, D. J. Barnes, S. T. Kay, I. G. McCarthy, D. Rennehan, and M. Vogelsberger. Is there enough star formation in simulated protoclusters? *MNRAS*, 501(2):1803–1822, Feb. 2021. doi: 10.1093/MNRAS/STAA3693.
- J. S. Mathis, W. Rumpl, and K. H. Nordsieck. The size distribution of interstellar grains. *ApJ*, 217: 425–433, Oct. 1977. doi: 10.1086/155591.
- L. Mattsson, A. Bhatnagar, F. A. Gent, and B. Villarroel. Clustering and dynamic decoupling of dust grains in turbulent molecular clouds. *MNRAS*, 483(4):5623–5641, Mar. 2019. doi: 10.1093/MNRAS/STY3369.
- C. McKee. Dust Destruction in the Interstellar Medium. In L. J. Allamandola and A. G. G. M. Tielens, editors, *Interstellar Dust*, volume 135 of *IAU Symposium*, page 431, 1989.
- G. A. D. Monteiro. *Estudos numéricos da viabilidade do dínamo de Tayler-Spruit em interiores estelares: o papel da rotação na instabilidade de Tayler*. PhD thesis, Universidade Federal de Minas Gerais, November 2024. URL [HTTPS://REPOSITORIO.UFMG.BR/BITSTREAMS/ACA29859-75E6-4418-B4E2-F798520AEFCC/DOWNLOAD](https://repositorio.ufmg.br/bitstreams/ACA29859-75E6-4418-B4E2-F798520AEFCC/download).

- M.-T. Nguyen, A.-D. Nguyen, J. Rantaharju, T. Puro, M. Rheinhardt, M. Korpi-Lagg, and H.-L. Truong. Supporting Opportunistic Data Operations for Data-Intensive Computational Applications . In *2024 IEEE International Conference on Big Data (BigData)*, pages 3735–3744, Los Alamitos, CA, USA, Dec. 2024. IEEE Computer Society. doi: 10.1109/BIGDATA62323.2024.10826079. URL [HTTPS://DOI.IEEECOMPUTERSOCIETY.ORG/10.1109/BIGDATA62323.2024.10826079](https://doi.ieeecomputersociety.org/10.1109/BIGDATA62323.2024.10826079).
- B. D. Oppenheimer, A. Babul, Y. Bahé, I. S. Butsky, and I. G. McCarthy. Simulating Groups and the IntraGroup Medium: The Surprisingly Complex and Rich Middle Ground between Clusters and Galaxies. *Universe*, 7(7):209, June 2021. doi: 10.3390/UNIVERSE7070209.
- J. Pekkilä, M. S. Väisälä, M. J. Käpylä, M. Rheinhardt, and O. Lappi. Scalable communication for high-order stencil computations using CUDA-aware MPI. *Parallel Computing*, 111:102904, July 2022. doi: 10.1016/J.PARCO.2022.102904.
- F. I. Pelupessy, A. van Elteren, N. de Vries, S. L. W. McMillan, N. Drost, and S. F. Portegies Zwart. The Astrophysical Multipurpose Software Environment. *A&A*, 557:A84, Sept. 2013. doi: 10.1051/0004-6361/201321252.
- Pencil Code Collaboration, A. Brandenburg, A. Johansen, P. Bourdin, W. Dobler, W. Lyra, M. Rheinhardt, S. Bingert, N. Haugen, A. Mee, F. Gent, N. Babkovskaia, C.-C. Yang, T. Heinemann, B. Dintrans, D. Mitra, S. Candelaresi, J. Warnecke, P. Käpylä, A. Schreiber, P. Chatterjee, M. Käpylä, X.-Y. Li, J. Krüger, J. Aarnes, G. Sarson, J. Oishi, J. Schober, R. Plasson, C. Sandin, E. Karchniwy, L. Rodrigues, A. Hubbard, G. Guerrero, A. Snodin, I. Losada, J. Pekkilä, and C. Qian. The Pencil Code, a modular MPI code for partial differential equations and particles: multipurpose and multiuser-maintained. *The Journal of Open Source Software*, 6(58):2807, Feb. 2021. doi: 10.21105/JOSS.02807.
- K. Rowlands, H. L. Gomez, L. Dunne, A. Aragón-Salamanca, S. Dye, S. Maddox, E. da Cunha, and P. van der Werf. The dust budget crisis in high-redshift submillimetre galaxies. *MNRAS*, 441: 1040–1058, June 2014. doi: 10.1093/MNRAS/STU605.
- V. Saeedzadeh, S. L. Jung, D. Rennehan, A. Babul, M. Tremmel, T. R. Quinn, Z. Shao, P. Sharma, L. Mayer, E. O'Sullivan, and S. I. Loubser. Cool and gusty, with a chance of rain: dynamics of multiphase CGM around massive galaxies in the Romulus simulations. *MNRAS*, 525(4):5677–5701, Nov. 2023. doi: 10.1093/MNRAS/STAD2637.
- J. C. Weingartner and B. T. Draine. Dust Grain-Size Distributions and Extinction in the Milky Way, Large Magellanic Cloud, and Small Magellanic Cloud. *ApJ*, 548:296–309, Feb. 2001. doi: 10.1086/318651.
- D. J. Whitworth, S. Srinivasan, R. E. Pudritz, M.-M. Mac Low, G. Eadie, A. Palau, J. D. Soler, R. J. Smith, K. Pattle, H. Robinson, R. Pillsworth, J. Wadsley, N. Brucy, U. Lebreuilly, P. Hennebelle, P. Girichidis, F. A. Gent, J. Marin, L. Sánchez Valido, V. Camacho, R. S. Klessen, and E. Vázquez-Semadeni. On the relation between magnetic field strength and gas density in the interstellar medium: a multiscale analysis. *MNRAS*, 540(3):2762–2786, July 2025. doi: 10.1093/MNRAS/STAF901.
- Y. Xia, H. Niu, Z. Zhang, H. Liu, and C. Wu. Extended multiscale isogeometric analysis for mechanical simulation of two-dimensional periodic heterogeneous materials. *Composite Structures*, 315: 116988, 2023. ISSN 0263-8223. doi: [HTTPS://DOI.ORG/10.1016/J.COMPSTRUCT.2023.116988](https://doi.org/10.1016/J.COMPSTRUCT.2023.116988). URL [HTTPS://WWW.SCIENCEDIRECT.COM/SCIENCE/ARTICLE/PII/S026382232300332X](https://www.sciencedirect.com/science/article/pii/S026382232300332X).